

Endpoint Regularization of Logarithmic Harmonic Polylogarithms

Stieltjes coefficients, Gamma transfer,
spectral finite parts, and exact antiderivatives

Research continuation prepared for Vladimir Reshetnikov
with OpenAI assistance
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Abstract

We give an explicit coefficient calculus for the endpoint constants of symmetrized, logarithmically decorated harmonic polylogarithms on a common Hurwitz lattice. A normally convergent canonical product expresses the sharp-cutoff polynomials through generalized Stieltjes constants and spectral derivatives of Hurwitz zeta. A proved Laplace–Stieltjes transfer converts these polynomials to Abel endpoint polynomials by the finite operator $\Gamma(1 + \partial_L)$. This yields formulas at every depth and every fixed logarithmic decoration, including a single-decoration family in which all positive-index Stieltjes constants cancel at the unshifted lattice. Explicit examples involve $\zeta'(2)$, $\zeta''(2)$, $\zeta'(3)$ and ordinary zeta products. We separately compute diagonal spectral finite parts and their ray-dependent correction terms: they are not the same regularization. A beta-weighted moment law evaluates the full unweighted family at every depth. Convergent repeated antiderivatives admit hypergeometric generators; at every positive integer shift a finite deletion formula reduces the entire unweighted hierarchy to the classical height-one Gamma quotient. Rational shifts give cyclotomic identities with an explicit logarithmic scale correction.

The symmetric-sum principle, Gamma regularization, and the height-one Gamma identity are classical ingredients, not claimed discoveries. The contribution is the unified, normalization-explicit shifted and decorated calculus, its consequences, and its reproducible implementation. All stated universal identities have proofs; finite symbolic tests and numerical diagnostics are kept distinct from those proofs. No arithmetic independence, resolution of the repository’s S_6 or S_8 conjectures, or proof-assistant formalization is claimed.

Repository snapshot:

`fc4d3bf80534ad7c901d3b8c9e71baf2df0064ed`

Complete source, validation scripts, recorded outputs and audit notes accompany this article.

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1 Research question, scope, and relation to existing work

The polylogarithm programme in ProveIt connects harmonic sums with Gamma functions, Hurwitz-zeta jets, generalized Stieltjes constants and integration operators. Its canonical manuscript already distinguishes proved reductions from numerical candidates and, in particular, retains the conjectural status of S_6 and S_8 [7]. We continue a different exact-identity question:

For a fully symmetrized harmonic sum with prescribed powers of logarithms, can one compute the entire endpoint subtraction polynomial, and hence its constant term, at arbitrary depth and a common positive Hurwitz shift? How does the answer change when the regularization is changed?

The answer is affirmative for the family defined below. The coefficients require only finitely many single-variable Hurwitz-zeta derivatives and Stieltjes constants. This is a statement about the explicitly symmetrized family, not a reduction theorem for every ordered multiple polylogarithm. The distinction is essential: symmetrization is the step that removes the unrestricted nested-order information.

There are three layers of results. First, Theorems 3.1 and 4.2 give all-depth cutoff and Abel polynomials. Second, Theorem 6.1 gives an exact generating function for one arbitrary logarithmic decoration and proves cancellation of positive-index Stieltjes constants at $a = 1$ in every depth greater than one. Third, Theorems 8.1, 9.1 and 9.3 give explicit regularization anomalies, all-depth beta moments and a convergent antiderivative hierarchy. The remaining results supply higher decorated examples, shift laws, differential recurrences and rational-lattice transport.

1.1 What is classical and what is developed here

The finite symmetric-sum identity is an elementary form of the symmetric multiple-zeta relations associated with Hoffman [2]; it can be proved without invoking multiple-zeta continuation. The comparison between harmonic and shuffle regularization is developed by Ihara, Kaneko and Zagier [3]. Kaneko and Sakata give a particularly useful formulation in terms of cutoff and polylogarithmic asymptotic polynomials [4]. Our cutoff variable is $\log N$, rather than $\log N + \gamma$. This choice accounts for the Euler-constant translation in the Gamma operator below.

The height-one generating identity

$$1 - \frac{\Gamma(1-u)\Gamma(1-v)}{\Gamma(1-u-v)} = \sum_{p,d \geq 1} \zeta(p+1, \{1\}^{d-1}) v^p u^d$$

is classical, following Aomoto and Drinfeld; see [5, 6, 4]. We give a beta-integral proof because its finite-shift extension requires careful normalization. We do not present the unshifted identity as new. Likewise, the Hurwitz-zeta and Gamma identities used in the proofs are standard [1].

The present continuation develops these ingredients simultaneously for arbitrary finite logarithmic content and a common positive shift, states the complete subtraction polynomials, compares them with diagonal and general-ray spectral constants, and makes the single-decoration cancellation explicit. These formulas are proved here, but no claim of worldwide priority is made. The literature and repository review was targeted rather than exhaustive.

1.2 Repository review boundary

The inspected material consists of the canonical manuscript's README and main driver; selected portions of its harmonic-depth and integration chapters; the log-Gamma-moment development; the alternating-harmonic continuation README; and the incoming directory listing and intake instructions. The alternating-harmonic continuation already treats $[u^r] \prod_{j \leq n} (1 + u/j)$ through incomplete-beta reflection [8]. Our subject is nonalternating endpoint regularization with independent logarithmic decorations, not a new proof of that reflection theorem.

The five incoming ZIP filenames were visible, but their binary payloads were not inspected. Consequently this article does not certify nonoverlap with those unexamined payloads, nor does it audit all 417 pages of the canonical manuscript. The provenance and audit files specify this boundary. No original repository file has been changed.

2 Definitions and the one-variable input

Throughout, $a > 0$ is real. Every logarithm of a lattice point $n + a$ is the real logarithm. We use the descending-index convention

$$\text{Li}_{s_1, \dots, s_d}(z_1, \dots, z_d) = \sum_{k_1 > \dots > k_d \geq 1} \frac{z_1^{k_1} \dots z_d^{k_d}}{k_1^{s_1} \dots k_d^{s_d}}. \quad (2.1)$$

The references using ascending indices must therefore be read with their indices reversed. When only $z_1 = z$ is displayed, the other arguments are one. For $|z| < 1$, the exponential damping in the largest index justifies every fixed spectral differentiation used here, locally uniformly in the orders.

Write

$$\zeta(1+t, a) = \frac{1}{t} + \sum_{r \geq 0} \frac{(-1)^r \gamma_r(a)}{r!} t^r, \quad \gamma_0(a) = -\psi(a), \quad \gamma = \gamma_0(1). \quad (2.2)$$

A prime on ζ always differentiates its first, spectral argument; ∂_a will always be written explicitly. Define

$$H_N^{(s)}(a) = \sum_{n=0}^{N-1} (n+a)^{-s}, \quad h_{r,N}(a) = \sum_{n=0}^{N-1} \frac{\log^r(n+a)}{n+a}. \quad (2.3)$$

Thus $h_{r,N} = (-1)^r \partial_s^r H_N^{(s)}(a)|_{s=1}$. At $a = 1$, $H_N^{(k)}(1)$ is the usual generalized harmonic number. We use the rising Pochhammer symbol $(a)_n = \Gamma(a+n)/\Gamma(a)$, with $(a)_0 = 1$.

Lemma 2.1 (Exact finite sums and cutoff constants). *For $r \geq 0$ and $N \geq 0$,*

$$h_{r,N}(a) = \gamma_r(a) - \gamma_r(N+a). \quad (2.4)$$

As $N \rightarrow \infty$,

$$h_{r,N}(a) = \frac{\log^{r+1}(N+a)}{r+1} + \gamma_r(a) + O_a\left(\frac{(1+\log N)^r}{N}\right). \quad (2.5)$$

For integers $k \geq 2$ and $R \geq 0$,

$$\sum_{n=0}^{N-1} \frac{\log^R(n+a)}{(n+a)^k} = (-1)^R (\zeta^{(R)}(k, a) - \zeta^{(R)}(k, N+a)), \quad (2.6)$$

$$\sum_{n \geq 0} \frac{\log^R(n+a)}{(n+a)^k} = (-1)^R \zeta^{(R)}(k, a). \quad (2.7)$$

All estimates are uniform when a stays in a compact subinterval of $(0, \infty)$.

Proof. Subtract the Laurent expansions in $\zeta(s, a) - \zeta(s, N + a) = H_N^{(s)}(a)$ and compare coefficients at $s = 1$. The poles cancel and give (2.4), with the sign in (2.2) unchanged.

For the asymptotic, the first Euler summation formula gives, locally uniformly for t near zero,

$$\zeta(1+t, x) = \frac{x^{-t}}{t} + \frac{1}{2}x^{-1-t} + (1+t) \int_0^\infty \frac{\tilde{B}_1(y)}{(x+y)^{2+t}} dy,$$

where $\tilde{B}_1(y) = 1/2 - \{y\}$; its values at integers are immaterial. Only boundedness is used. Differentiation of the integral r times produces an error $O(x^{-1}(1+\log x)^r)$; this follows by putting $y = xv$ and integrating $(1+v)^{-2}$ times a fixed power of $\log(1+v)$. The constant term removed from x^{-t}/t has coefficient $(-1)^{r+1} \log^{r+1} x / (r+1)!$. Comparison with (2.2) yields

$$\gamma_r(x) = -\frac{\log^{r+1} x}{r+1} + O(x^{-1}(1+\log x)^r).$$

Together with (2.4), this proves (2.5). The remaining identities follow from the absolutely convergent Hurwitz series and spectral differentiation. The tail is $O(N^{1-k}(1+\log N)^R)$ by an integral comparison. Uniformity in a follows from the same estimates. $\square$

For integer $k \geq 2$ the undecorated version is also a polygamma identity:

$$H_N^{(k)}(a) = \frac{(-1)^k}{(k-1)!} (\psi^{(k-1)}(a) - \psi^{(k-1)}(N+a)). \quad (2.8)$$

This is one direct bridge between the generalized harmonic numbers and the polygamma functions in the construction.

Fix an integer $R \geq 0$. Put

$$f_r(x) = \frac{\log^r x}{x}, \quad W_{\mathbf{u}}(x) = \sum_{r=0}^R u_r f_r(x), \quad V_{\mathbf{u}}(L) = \sum_{r=0}^R \frac{u_r L^{r+1}}{r+1}.$$

A *content* is $\mathbf{m} = (m_0, \dots, m_R) \in \mathbb{N}_0^{R+1}$. Its depth, logarithmic degree and total degree are

$$d = |\mathbf{m}| = \sum_r m_r, \quad w(\mathbf{m}) = \sum_r r m_r, \quad D(\mathbf{m}) = d + w(\mathbf{m}) = \sum_r (r+1) m_r. \quad (2.9)$$

Let $\mathbf{u}^{\mathbf{m}} = \prod_r u_r^{m_r}$ and define

$$e_{\mathbf{m}, N}(a) = [\mathbf{u}^{\mathbf{m}}] \prod_{n=0}^{N-1} (1 + W_{\mathbf{u}}(n+a)). \quad (2.10)$$

For $d > 0$, this is a sum over strictly ordered lattice points and distinct words of logarithmic powers with content $\mathbf{m}$. Repeated decorations do *not* introduce an extra factorial. For example,

$$e_{(1,1), N}(a) = \sum_{N > n > m \geq 0} \frac{\log(n+a) + \log(m+a)}{(n+a)(m+a)},$$

whereas $e_{(0,2), N}$ has numerator $\log(n+a) \log(m+a)$, not twice that numerator.

For $0 < z < 1$ define the corresponding Abel sum

$$E_{\mathbf{m}}(z; a) = \sum_{n_1 > \dots > n_d \geq 0} z^{n_1+a} \sum_{\text{cont}(r_1, \dots, r_d) = \mathbf{m}} \prod_{i=1}^d f_{r_i}(n_i + a). \quad (2.11)$$

There is one summand for each distinct word of the indicated content. Define $E_0 = 1$ separately. At $a = 1$,

$$E_{\mathbf{m}}(z; 1) = (-1)^{w(\mathbf{m})} \sum_{\text{cont}(\mathbf{r})=\mathbf{m}} \partial_{s_1}^{r_1} \cdots \partial_{s_d}^{r_d} \text{Li}_{s_1, \dots, s_d}(z, 1, \dots, 1) \Big|_{s_1=\dots=s_d=1}. \quad (2.12)$$

This places the entire coefficient calculus inside spectral derivatives of multiple polylogarithms, rather than merely formal harmonic notation.

3 The all-depth cutoff coefficient theorem

Define a canonical product in the $R + 1$ auxiliary variables:

$$\mathcal{C}_a(\mathbf{u}) = \exp\left(\sum_{r=0}^R \gamma_r(a) u_r\right) \prod_{n \geq 0} (1 + W_{\mathbf{u}}(n + a)) e^{-W_{\mathbf{u}}(n+a)}. \quad (3.1)$$

This is an ordinary entire function of finitely many complex variables. It must not be confused with the formal Gamma-moment series introduced later.

Theorem 3.1 (Cutoff polynomials and exact coefficient closure). *The product (3.1) converges normally on every compact subset of $\mathbb{C}^{R+1}$. Near $\mathbf{u} = 0$, its logarithm is*

$$\begin{aligned} \log \mathcal{C}_a(\mathbf{u}) &= \sum_{r=0}^R \gamma_r(a) u_r \\ &+ \sum_{k \geq 2} \frac{(-1)^{k-1}}{k} \sum_{r_1, \dots, r_k=0}^R (-1)^{r_1 + \dots + r_k} \zeta^{(r_1 + \dots + r_k)}(k, a) u_{r_1} \cdots u_{r_k}. \end{aligned} \quad (3.2)$$

For every content $\mathbf{m}$, set

$$P_{\mathbf{m},a}(L) = [\mathbf{u}^{\mathbf{m}}] \mathcal{C}_a(\mathbf{u}) e^{V_{\mathbf{u}}(L)}. \quad (3.3)$$

If $d > 0$, this is a polynomial of degree exactly $D(\mathbf{m})$, with leading coefficient

$$\frac{1}{\prod_{r=0}^R m_r! (r+1)^{m_r}}.$$

Moreover,

$$e_{\mathbf{m},N}(a) = P_{\mathbf{m},a}(\log(N + a)) + O_{a,\mathbf{m}}\left(\frac{(1 + \log N)^{D(\mathbf{m})}}{N}\right). \quad (3.4)$$

In particular, the sharp-cutoff constant is

$$C_{\mathbf{m}}(a) := P_{\mathbf{m},a}(0) = [\mathbf{u}^{\mathbf{m}}] \mathcal{C}_a(\mathbf{u}). \quad (3.5)$$

Every coefficient of $P_{\mathbf{m},a}$ is a rational polynomial in the finitely many atoms $\gamma_r(a)$ and $\zeta^{(j)}(k, a)$, with $k \leq d$ and $j \leq w(\mathbf{m})$; only decorations present in $\mathbf{m}$ supply Stieltjes atoms.

Proof. On a compact set of auxiliary variables, $|W_{\mathbf{u}}(n+a)| \leq C(1 + \log n)^R/n$ for all sufficiently large n . Consequently $\sum_n |W_{\mathbf{u}}(n+a)|^2$ converges uniformly there. For the tail of the product, $\log(1+W) - W = O(|W|^2)$ uniformly; finitely many initial factors are entire. This proves normal convergence and entire dependence, including at zeros of individual factors. A sufficiently small polydisc around zero avoids all those zeros, so expanding its logarithm is legitimate. The power-sum identity (2.7) then proves (3.2).

For the asymptotic, take the logarithm of the *finite* product as a formal power series and retain total degree at most d :

$$\log \prod_{n < N} (1 + W_{\mathbf{u}}(n + a)) = \sum_{k=1}^d \frac{(-1)^{k-1}}{k} \sum_{n < N} W_{\mathbf{u}}(n + a)^k \pmod{\text{total degree} > d}. \quad (3.6)$$

The $k = 1$ coefficient is supplied by (2.5); it equals $V_{\mathbf{u}}(\log(N + a)) + \sum_r \gamma_r(a) u_r$ plus the stated errors. For $k \geq 2$ the relevant finite power sum differs from its limit by $O(N^{1-k}(1 + \log N)^{r_1 + \dots + r_k})$.

Exponentiation and extraction of a fixed content use only finitely many products. The other factors in such a product have at most the remaining logarithmic degree. A singleton error contributes at most a constant times $N^{-1}(1 + \log N)^{D-1}$; a block of size $k \geq 2$ contributes at most $N^{1-k}(1 + \log N)^{D-k}$. The weaker common error in (3.4) is therefore valid. This reasoning is entirely coefficientwise and does not assert uniform asymptotics for unrestricted auxiliary variables.

The highest power of L comes only from selecting all decorations in $e^{V_{\mathbf{u}}(L)}$, with the constant term of C_a . This gives the displayed leading coefficient. The finite atom statement follows directly from (3.2). The empty content gives $P_{\mathbf{0},a} = 1$ exactly. $\square$

3.1 First exact cutoff formulas

For compactness in examples, write $g_r = \gamma_r(a)$ and $Z_k^{[j]} = \zeta^{(j)}(k, a)$. With only u_0 present,

$$\begin{aligned} C_{(1)}(a) &= g_0, \\ C_{(2)}(a) &= \frac{g_0^2 - Z_2^{[0]}}{2}, \\ C_{(3)}(a) &= \frac{g_0^3 - 3g_0 Z_2^{[0]} + 2Z_3^{[0]}}{6}. \end{aligned} \quad (3.7)$$

The mixed logarithmic coefficients begin with

$$P_{(1,1),a}(L) = (L + g_0) \left(\frac{L^2}{2} + g_1 \right) + Z_2^{[1]}, \quad (3.8)$$

$$P_{(0,2),a}(L) = \frac{1}{2} \left(\frac{L^2}{2} + g_1 \right)^2 - \frac{1}{2} Z_2^{[2]}. \quad (3.9)$$

For instance, the plus sign before $Z_2^{[1]}$ in (3.8) comes from subtracting the diagonal sum $\sum \log(n + a)/(n + a)^2 = -\zeta'(2, a)$. This sign is easy to lose when a logarithmic insertion is identified with an unsigned spectral derivative.

If one assigns formal degree $r + 1$ to g_r , degree $k + j$ to $Z_k^{[j]}$, and degree one to L , every term in $P_{\mathbf{m},a}$ has degree $D(\mathbf{m})$. This is useful bookkeeping for an implementation. It is not a statement about a motivic weight filtration or arithmetic independence.

3.2 Why symmetrization matters

For labelled variables $x_{i,n}$, inclusion–exclusion on coincidences gives

$$\sum_{\substack{n_1, \dots, n_d \geq 0 \\ n_i \text{ distinct}}} \prod_{i=1}^d x_{i,n_i} = \sum_{\pi \in \Pi_d} \mu(\pi) \prod_{B \in \pi} \left(\sum_{n \geq 0} \prod_{i \in B} x_{i,n} \right), \quad (3.10)$$

when the sums are finite or absolutely convergent. Here Π_d is the set of set partitions and

$$\mu(\pi) = \prod_{B \in \pi} (-1)^{|B|-1} (|B| - 1)!.$$

One proof is to fix an equality pattern of the indices. Its coefficient on the right is the sum of the partition-lattice Möbius function below that pattern, hence is one for the discrete pattern and zero otherwise. An alternative elementary proof expands the finite logarithmic product in (3.6) and uses the exponential formula for set partitions.

With $x_{i,n} = f_{r_i}(n+a)$, the labelled distinct-index sum is $\prod_r m_r!$ times the content coefficient. This accounts for the absence of any additional factorial in (2.10). An individual ordered word has no such general reduction. We never infer its reduction from that of the complete symmetrization.

4 Gamma transfer from cutoff to Abel endpoints

Set $\epsilon = -\log z > 0$ and $L = \log(1/\epsilon)$. There is no ambiguity in these real variables. The more customary $L_z = -\log(1-z)$ satisfies $L_z = L + O(\epsilon)$, so replacing one by the other changes a fixed polynomial by $o(1)$ as $z \uparrow 1$.

Lemma 4.1 (Polynomial Laplace–Stieltjes transfer). *Let $S(x)$ be a step function with locally finite variation, supported in $[a, \infty)$ for some $a > 0$. Suppose that, as $x \rightarrow \infty$,*

$$S(x) = P(\log x) + O(x^{-1}(1 + \log x)^D)$$

for a polynomial P . Interpret the Stieltjes integral below as an improper integral if its total variation is not integrable. Then

$$\int_0^\infty e^{-\epsilon x} dS(x) = \sum_{j=0}^{\deg P} \frac{\Gamma^{(j)}(1)}{j!} P^{(j)}(L) + O(\epsilon(1+L)^{D+1}), \quad (4.1)$$

provided $D \geq \deg P$; an $o(1)$ remainder suffices without recording its rate.

Proof. Integration by parts gives $\epsilon \int_0^\infty e^{-\epsilon x} S(x) dx$. The boundary terms vanish because $S = 0$ near zero and grows at most polynomially in $\log x$. Replacing S by $P(\log x)$ on a bounded interval costs $O(\epsilon)$, since powers of $\log x$ are integrable at zero. On $x \geq A > 1$, the error is bounded by

$$C\epsilon \int_A^\infty e^{-\epsilon x} \frac{(1 + \log x)^D}{x} dx = O(\epsilon(1+L)^{D+1}).$$

Split this integral at $x = 1/\epsilon$ to obtain the last estimate. For the main term put $t = \epsilon x$. The polynomial Taylor identity and the Gamma integral give

$$\begin{aligned} \int_0^\infty e^{-t} P(L + \log t) dt &= \sum_{j=0}^{\deg P} \frac{P^{(j)}(L)}{j!} \int_0^\infty e^{-t} \log^j t dt \\ &= \sum_{j=0}^{\deg P} \frac{\Gamma^{(j)}(1)}{j!} P^{(j)}(L). \end{aligned}$$

Only a finite sum is interchanged with the integral. □

Theorem 4.2 (All-depth Abel endpoint polynomial). *For every content $\mathbf{m}$, define*

$$Q_{\mathbf{m},a}(L) = \Gamma(1 + \partial_L)P_{\mathbf{m},a}(L) := \sum_{j=0}^{D(\mathbf{m})} \frac{\Gamma^{(j)}(1)}{j!} \partial_L^j P_{\mathbf{m},a}(L). \quad (4.2)$$

Then, for $d > 0$,

$$E_{\mathbf{m}}(e^{-\epsilon}; a) = Q_{\mathbf{m},a}(L) + O_{a,\mathbf{m}}(\epsilon(1+L)^{D(\mathbf{m})+1}). \quad (4.3)$$

The Abel endpoint constant is the ordinary limit after polynomial subtraction:

$$A_{\mathbf{m}}(a) := Q_{\mathbf{m},a}(0) = \lim_{z \uparrow 1} (E_{\mathbf{m}}(z; a) - Q_{\mathbf{m},a}(-\log(1-z)) + Q_{\mathbf{m},a}(0)). \quad (4.4)$$

It belongs to the algebra of Theorem 3.1, enlarged only by γ and ordinary zeta values $\zeta(2), \dots, \zeta(D(\mathbf{m}))$.

Proof. Let $S(x)$ sum the finite products over tuples whose largest lattice point is at most x . Its jumps are exactly the coefficients of the Abel series in (2.11). Between lattice points, $S(x) = e_{\mathbf{m},N(x)}(a)$ with $N(x) + a = x + O(1)$. Thus Theorem 3.1 gives the hypothesis of Lemma 4.1. Applying that lemma proves (4.3). Replacing L by $-\log(1-z)$ changes only the vanishing error, proving the limit statement. Finally,

$$\log \Gamma(1+u) = -\gamma u + \sum_{k \geq 2} \frac{(-1)^k \zeta(k)}{k} u^k \quad (4.5)$$

shows that each $\Gamma^{(j)}(1)$ is a rational polynomial in the asserted ordinary constants. The empty content is again handled by definition. $\square$

The first Gamma moments are

$$\begin{aligned} \Gamma'(1) &= -\gamma, \\ \Gamma''(1) &= \gamma^2 + \zeta(2), \\ \Gamma'''(1) &= -\gamma^3 - 3\gamma\zeta(2) - 2\zeta(3), \\ \Gamma^{(4)}(1) &= \gamma^4 + 6\gamma^2\zeta(2) + 3\zeta(2)^2 + 8\gamma\zeta(3) + 6\zeta(4). \end{aligned} \quad (4.6)$$

These formulas explain exactly which new ordinary constants an Abel endpoint can introduce. No new positive-index generalized Stieltjes constant is created by the transfer itself.

4.1 A formal master series, and a necessary convergence warning

Define, as a formal multivariate power series,

$$\mathcal{K}(\mathbf{u}) = \sum_{\mathbf{m} \in \mathbb{N}_0^{R+1}} \frac{\Gamma^{(D(\mathbf{m}))}(1)}{\prod_{r=0}^R m_r! (r+1)^{m_r}} \mathbf{u}^{\mathbf{m}}. \quad (4.7)$$

Theorem 4.2 gives the compact master identity

$$\boxed{\sum_{\mathbf{m}} A_{\mathbf{m}}(a) \mathbf{u}^{\mathbf{m}} = \mathcal{C}_a(\mathbf{u}) \mathcal{K}(\mathbf{u})}. \quad (4.8)$$

Indeed, the coefficient of $\mathbf{u}^{\mathbf{m}}$ in $e^{V_{\mathbf{u}}(\log t)}$ integrates against $e^{-t} dt$ to the coefficient in (4.7). For a fixed content this is a finite, valid integration of logarithmic moments.

Proposition 4.3 (The Gamma-moment master series need not converge). *When a u_1 variable is present, $\mathcal{K}(0, u_1, 0, \dots)$ has radius of convergence zero. Consequently (4.8) is not, in general, an identity of analytic germs in all logarithmic-decoration variables.*

Proof. The coefficient of u_1^m is $\Gamma^{(2m)}(1)/(2^m m!)$. All integrands in

$$\Gamma^{(2m)}(1) = \int_0^\infty e^{-t} \log^{2m} t \, dt \geq e^{-1} \int_0^1 \log^{2m} t \, dt = e^{-1} (2m)!$$

are nonnegative. The m th roots of $(2m)!/(2^m m!)$ tend to infinity. This proves the assertion. In particular, the tempting expression $\int_0^\infty e^{-t} \exp(u_1 \log^2 t/2) \, dt$ diverges near $t = 0$ for every real $u_1 > 0$. $\square$

This warning does not affect any fixed coefficient, nor the analytic one-decoration generators below. It does prevent an invalid application of holomorphic parameter differentiation to the unrestricted master integral. The entire canonical product $\mathcal{C}_a$ and the formal moment series $\mathcal{K}$ have fundamentally different convergence properties.

4.2 A Gamma–polygamma differential recurrence

On polynomials in L , put $D_L = \partial_L$, $\mathcal{G} = \Gamma(1 + D_L)$, and

$$\mathcal{T} = L + \psi(1 + D_L) = L - \gamma + \sum_{k \geq 1} (-1)^{k+1} \zeta(k+1) D_L^k. \quad (4.9)$$

All operator series terminate on a fixed polynomial.

Proposition 4.4 (Closed recurrence for endpoint polynomials). *With a negative content interpreted as zero,*

$$\partial_L Q_{\mathbf{m},a}(L) = \sum_{r=0}^R \mathcal{T}^r Q_{\mathbf{m}-\mathbf{e}_r,a}(L). \quad (4.10)$$

The powers of $\mathcal{T}$ mean composition of the displayed operator, not ordinary commutative powers of its separate summands.

Proof. The relation $[D_L, L] = 1$ gives $\mathcal{G}L\mathcal{G}^{-1} = L + \psi(1 + D_L)$. This follows first for a formal series $G(D_L)$ from $G(D_L)L = LG(D_L) + G'(D_L)$; the constant term of G is one here, so its inverse exists on polynomials. Hence $\mathcal{G}L^r\mathcal{G}^{-1} = \mathcal{T}^r$. Differentiating $\mathcal{C}_a(\mathbf{u})e^{V_{\mathbf{u}}(L)}$ gives $\partial_L P_{\mathbf{m},a} = \sum_r L^r P_{\mathbf{m}-\mathbf{e}_r,a}$. Conjugate this equality by $\mathcal{G}$, which commutes with ∂_L , to obtain (4.10). $\square$

5 Gamma quotients, common shifts, and parameter differentiation

When $u_0 = u$ and the other auxiliary variables vanish, the canonical product collapses to an inverse Gamma quotient:

$$\mathcal{C}_a(u, 0, \dots) = \frac{\Gamma(a)}{\Gamma(a+u)}. \quad (5.1)$$

To see this, compare the logarithms near zero. The linear coefficient is $-\psi(a)$, and the coefficient of u^k for $k \geq 2$ is $(-1)^{k-1} \zeta(k, a)/k$, in agreement with (3.2). Since both sides are entire functions of u , the local identity proves the formula everywhere.

Corollary 5.1 (Unweighted endpoint generator). *For the depth- d content (d),*

$$\begin{aligned} \sum_{d \geq 0} P_{(d),a}(L) u^d &= \frac{\Gamma(a)}{\Gamma(a+u)} e^{uL}, \\ \sum_{d \geq 0} Q_{(d),a}(L) u^d &= \mathcal{A}_a(u) e^{uL}, \quad \mathcal{A}_a(u) := \frac{\Gamma(a)\Gamma(1+u)}{\Gamma(a+u)}. \end{aligned} \quad (5.2)$$

In particular, $A_{(d)}(a) = [u^d] \mathcal{A}_a(u)$. At $a = 1$,

$$Q_{(d),1}(L) = \frac{L^d}{d!}, \quad A_{(d)}(1) = 0 \quad (d \geq 1). \quad (5.3)$$

These endpoint identities have an independent full-function check. Let $\mathcal{E}(z, u; a) = 1 + \sum_{d \geq 1} u^d E_{(d)}(z; a)$.

Proposition 5.2 (An exact incomplete-beta generator). *For $a > 0$, $0 < z < 1$, and u near zero,*

$$\mathcal{E}(z, u; a) = 1 + \frac{uz^a}{a} {}_2F_1(1, a+u; a+1; z) \quad (5.4)$$

$$= 1 + u(1-z)^{-u} B_z(a, u), \quad (5.5)$$

where $B_z(a, u) = \int_0^z t^{a-1} (1-t)^{u-1} dt$. As $z \uparrow 1$, locally uniformly in a small complex disc about $u = 0$,

$$\mathcal{E}(z, u; a) = \mathcal{A}_a(u) (1-z)^{-u} + O(1-z). \quad (5.6)$$

Proof. Condition on the largest index n . Its contribution to the generating function is

$$\frac{uz^{n+a}}{n+a} \prod_{j=0}^{n-1} \left(1 + \frac{u}{a+j} \right) = \frac{uz^a}{a} \frac{(a+u)_n}{(a+1)_n} z^n.$$

Since $(1)_n/n! = 1$, summing gives (5.4). Euler's hypergeometric transformation gives (5.5); alternatively expand its integral into a binomial series. The incomplete integral is entire in u for fixed $z < 1$, so the identity is not limited to $\operatorname{Re} u > 0$.

Initially for $\operatorname{Re} u > 0$, write $B_z(a, u) = B(a, u) - B_{1-z}(u, a)$. The complete beta term is $uB(a, u) = \mathcal{A}_a(u)$. Taylor expansion of $(1-t)^{a-1}$ in the complementary integral gives

$$u(1-z)^{-u} B_{1-z}(u, a) = 1 + O(1-z).$$

The coefficients in this last expansion extend holomorphically through $u = 0$, since their only possible denominators are $u+j$ with $j \geq 1$ after the constant term. This proves uniformity near zero and (5.6). $\square$

At the half shift,

$$\log \mathcal{A}_{1/2}(u) = 2 \log 2 u + \sum_{k \geq 2} \frac{(-1)^k (2-2^k) \zeta(k)}{k} u^k. \quad (5.7)$$

Consequently

$$\begin{aligned} A_{(1)}(1/2) &= 2 \log 2, \\ A_{(2)}(1/2) &= 2 \log^2 2 - \zeta(2), \\ A_{(3)}(1/2) &= \frac{4}{3} \log^3 2 - 2\zeta(2) \log 2 + 2\zeta(3). \end{aligned} \quad (5.8)$$

For a positive integer M ,

$$\mathcal{A}_M(u) = \prod_{j=1}^{M-1} \left(1 + \frac{u}{j}\right)^{-1}. \quad (5.9)$$

Its coefficient of u^d is $(-1)^d$ times the degree- d complete homogeneous symmetric function of the reciprocals of the positive integers smaller than M . Equivalently,

$$A_{(d)}(M) = \frac{(-1)^d}{d!} Y_d(H_{M-1}, 1!H_{M-1}^{(2)}, \dots, (d-1)!H_{M-1}^{(d)}), \quad (5.10)$$

where the complete exponential Bell polynomials satisfy

$$Y_d(x_1, \dots, x_d) = d![t^d] \exp \left(\sum_{j=1}^d \frac{x_j t^j}{j!} \right).$$

Thus this specialization gives finite generalized-harmonic identities at all depths, not merely a collection of isolated low-order constants.

5.1 Exact shift and derivative laws

Proposition 5.3 (Deletion of the first lattice point). *As entire functions, with removable quotients interpreted by cancellation,*

$$\mathcal{C}_{a+1}(\mathbf{u}) = \frac{\mathcal{C}_a(\mathbf{u})}{1 + W_{\mathbf{u}}(a)}. \quad (5.11)$$

The same law holds for the formal Abel-constant generator in (4.8). In coefficient form,

$$A_{\mathbf{m}}(a+1) + \sum_{r=0}^R f_r(a) A_{\mathbf{m}-\mathbf{e}_r}(a+1) = A_{\mathbf{m}}(a). \quad (5.12)$$

Proof. The Hurwitz recurrence gives $\gamma_r(a+1) = \gamma_r(a) - f_r(a)$. Remove the $n=0$ factor from (3.1); its exponential compensation cancels this Stieltjes shift exactly. This proves (5.11). The formal factor $\mathcal{K}$ is independent of a , so multiplication by it proves the Abel law. Extraction of a content gives (5.12). $\square$

The elementary Hurwitz differential identity $\partial_a \zeta(s, a) = -s\zeta(s+1, a)$ yields

$$\boxed{\gamma'_r(a) = (-1)^{r+1} (\zeta^{(r)}(2, a) + r\zeta^{(r-1)}(2, a))}, \quad (5.13)$$

where the second term is omitted for $r=0$. This follows by comparing the coefficient of t^r in $-(1+t)\zeta(2+t, a)$ and (2.2). In particular,

$$\gamma'_0(a) = -\zeta(2, a), \quad \gamma'_1(a) = \zeta'(2, a) + \zeta(2, a).$$

Higher parameter derivatives use

$$\partial_a^q \zeta(s, a) = (-1)^q (s)_q \zeta(s+q, a). \quad (5.14)$$

These identities differentiate every finite coefficient formula in this article, producing further exact polygamma and zeta-derivative identities. They do not confuse spectral differentiation with differentiation of the Hurwitz parameter. Normal convergence of the canonical product on a small complex neighborhood of a positive a also justifies its parameter differentiation directly.

6 One arbitrary logarithmic decoration at every depth

Let $\ell \geq 1$. Write $P_{r;\ell,a}$, $Q_{r;\ell,a}$ and $A_{r;\ell}(a)$ for the content with $m_0 = r$, $m_\ell = 1$, and all other entries zero. Here the total depth is $r + 1$, not r . For instance,

$$E_{r;\ell}(z; 1) = \sum_{n_1 > \dots > n_{r+1} \geq 1} \frac{z^{n_1}}{n_1 \cdots n_{r+1}} \sum_{i=1}^{r+1} \log^\ell n_i.$$

Define the analytic germ

$$b_{a,\ell}(u) = \gamma_\ell(a) + \sum_{j \geq 1} (-1)^{j+\ell} \zeta^{(\ell)}(j+1, a) u^j \quad (6.1)$$

$$= \gamma_\ell(a) - u \sum_{n \geq 0} \frac{\log^\ell(n+a)}{(n+a)(n+a+u)}. \quad (6.2)$$

The series in (6.1) converges for $|u| < a$; the normally convergent resolvent sum gives its meromorphic continuation away from the indicated lattice poles.

Theorem 6.1 (Single-decoration identity and Stieltjes cancellation). *For u sufficiently close to zero,*

$$\begin{aligned} \sum_{r \geq 0} Q_{r;\ell,a}(L) u^r &= \frac{\Gamma(a)}{\Gamma(a+u)} \left\{ b_{a,\ell}(u) \Gamma(1+u) e^{uL} \right. \\ &\quad \left. + \frac{1}{\ell+1} \partial_u^{\ell+1} (\Gamma(1+u) e^{uL}) \right\}. \end{aligned} \quad (6.3)$$

The derivative in braces acts only on its displayed Gamma-exponential factor. In particular,

$$\boxed{\sum_{r \geq 0} A_{r;\ell}(a) u^r = \mathcal{A}_a(u) \left(b_{a,\ell}(u) + \frac{\Gamma^{(\ell+1)}(1+u)}{(\ell+1)\Gamma(1+u)} \right)}. \quad (6.4)$$

At $a = 1$, for every $r \geq 1$,

$$\boxed{A_{r;\ell}(1) = (-1)^{r+\ell} \zeta^{(\ell)}(r+1) + \frac{1}{\ell+1} [u^r] \frac{\Gamma^{(\ell+1)}(1+u)}{\Gamma(1+u)}}. \quad (6.5)$$

Thus no positive-index Stieltjes constant occurs in this exact expression. The depth-one exception is

$$A_{0;\ell}(1) = \gamma_\ell + \frac{\Gamma^{(\ell+1)}(1)}{\ell+1}. \quad (6.6)$$

Proof. Use u for the undecorated variable and v for the single ℓ th logarithm. Differentiation of the logarithm of the canonical product at $v = 0$ gives

$$\begin{aligned} \partial_v \log \mathcal{C}_a(u, v)|_{v=0} &= \gamma_\ell(a) + \sum_{n \geq 0} f_\ell(n+a) \left(\frac{1}{1+u/(n+a)} - 1 \right) \\ &= b_{a,\ell}(u). \end{aligned}$$

This differentiation is justified by normal convergence of the compensated series. Expanding the resolvent proves (6.1), including its sign. The coefficient of v in the cutoff generating function is therefore

$$\frac{\Gamma(a)}{\Gamma(a+u)} e^{uL} \left(b_{a,\ell}(u) + \frac{L^{\ell+1}}{\ell+1} \right).$$

Applying $\Gamma(1 + \partial_L)$ and using $L^{\ell+1}e^{uL} = \partial_u^{\ell+1}e^{uL}$ proves (6.3). The prefactor $\Gamma(a)/\Gamma(a+u)$ is not differentiated in this operation. The equality can be read coefficientwise; equivalently the Gamma integral is convergent for u in a small disc, with only a fixed logarithmic power inserted. Setting $L = 0$ proves (6.4).

For $a = 1$ the prefactor $\mathcal{A}_1(u)$ equals one exactly. The only positive-index Stieltjes atom in $b_{1,\ell}$ is its constant term γ_ℓ . It cannot contribute to $[u^r]$ when $r \geq 1$, giving (6.5). The remaining Gamma quotient is the complete Bell polynomial

$$\frac{\Gamma^{(\ell+1)}(1+u)}{\Gamma(1+u)} = Y_{\ell+1}(\psi(1+u), \psi'(1+u), \dots, \psi^{(\ell)}(1+u)).$$

The Taylor coefficients of these polygamma functions involve only γ and ordinary zeta values. This proves the asserted cancellation and the depth-one exception. $\square$

The cancellation is an algebraic consequence of the Gamma quotient, not an assertion that any Stieltjes constant is independent of the surviving constants. It is also stronger than a finite numerical observation: both the depth $r+1$ and the logarithmic power ℓ are unrestricted.

6.1 The single-logarithm family in ordinary zeta notation

For $\ell = 1$, equations (6.3)–(6.4) become

$$\sum_{r \geq 0} Q_{r;1,a}(L)u^r = \mathcal{A}_a(u)e^{uL} \left(b_{a,1}(u) + \frac{(L + \psi(1+u))^2 + \psi'(1+u)}{2} \right), \quad (6.7)$$

$$\sum_{r \geq 0} A_{r;1}(a)u^r = \mathcal{A}_a(u) \left(b_{a,1}(u) + \frac{\psi(1+u)^2 + \psi'(1+u)}{2} \right). \quad (6.8)$$

Corollary 6.2 (An explicit all-depth zeta-derivative identity). *For every integer $r \geq 1$,*

$$A_{r;1}(1) = (-1)^r \left(-\zeta'(r+1) + \gamma\zeta(r+1) + \frac{r+1}{2}\zeta(r+2) + \frac{1}{2} \sum_{j=1}^{r-1} \zeta(j+1)\zeta(r-j+1) \right). \quad (6.9)$$

The sum is empty for $r = 1$.

Proof. In a disc about zero,

$$\psi(1+u) = -\gamma + \sum_{j \geq 1} (-1)^{j+1} \zeta(j+1)u^j, \quad \psi'(1+u) = \sum_{j \geq 0} (-1)^j (j+1) \zeta(j+2)u^j.$$

Multiply the first series by itself, add the second, divide by two, and extract $[u^r]$. Add the coefficient $(-1)^{r+1}\zeta'(r+1)$ from $b_{1,1}$. The result is (6.9). $\square$

The first three positive-depth-index constants are

$$A_{1;1}(1) = \zeta'(2) - \gamma\zeta(2) - \zeta(3), \quad (6.10)$$

$$A_{2;1}(1) = -\zeta'(3) + \gamma\zeta(3) + \frac{11}{4}\zeta(4), \quad (6.11)$$

$$A_{3;1}(1) = \zeta'(4) - \gamma\zeta(4) - 2\zeta(5) - \zeta(2)\zeta(3). \quad (6.12)$$

Here $\zeta(2)^2 = 5\zeta(4)/2$ was used in (6.11).

For a completely explicit ordinary limit, put

$$F(z) = \sum_{n>m \geq 1} \frac{z^n (\log n + \log m)}{nm}, \quad L_z = -\log(1-z).$$

Then Theorem 6.1 gives

$$\lim_{z \uparrow 1} \left[F(z) - \frac{L_z^3}{2} + \gamma L_z^2 - \left(\gamma_1 + \frac{\gamma^2}{2} + \frac{3\zeta(2)}{2} \right) L_z \right] = \zeta'(2) - \gamma\zeta(2) - \zeta(3). \quad (6.13)$$

Since $F(z) = -(\partial_{s_1} + \partial_{s_2}) \text{Li}_{s_1, s_2}(z, 1)|_{(1,1)}$, this is an exact endpoint identity for polylogarithmic order derivatives. The constant on the right is not obtained by assigning a value directly to the divergent series $F(1)$.

For a higher logarithmic power, the same theorem yields

$$A_{1;2}(1) = -\zeta''(2) + \gamma^2\zeta(2) + 2\gamma\zeta(3) + \frac{9}{2}\zeta(4). \quad (6.14)$$

The corresponding numerator is $\log^2 n + \log^2 m$, not $\log n \log m$. To verify (6.14) directly from the generating formula, differentiate $\Gamma'''/\Gamma = \psi^3 + 3\psi\psi' + \psi''$ once at zero, divide by three, and use (6.5). This produces $\gamma^2\zeta(2) + \zeta(2)^2 + 2\gamma\zeta(3) + 2\zeta(4)$, which is the stated expression.

7 Two logarithmic insertions and higher content

The notation $A_{(r,2)}$ in this section means r undecorated factors and *two* factors carrying one logarithm each. It differs from $A_{r;2}$, which denotes one factor carrying a squared logarithm. Set $b_a = b_{a,1}$ and

$$M_a(u) = -\sum_{n \geq 0} \frac{\log^2(n+a)}{(n+a+u)^2} = \sum_{j \geq 0} (-1)^{j+1} (j+1) \zeta''(j+2, a) u^j. \quad (7.1)$$

The first sum is normally convergent off its lattice poles, and the Taylor series is valid for $|u| < a$.

Proposition 7.1 (Two-logarithm constant generator). *For u near zero,*

$$\begin{aligned} \sum_{r \geq 0} A_{(r,2)}(a) u^r = \mathcal{A}_a(u) & \left[\frac{b_a(u)^2 + M_a(u)}{2} + \frac{b_a(u) \Gamma''(1+u)}{2\Gamma(1+u)} \right. \\ & \left. + \frac{\Gamma^{(4)}(1+u)}{8\Gamma(1+u)} \right]. \end{aligned} \quad (7.2)$$

Proof. The first and second derivatives of $\log \mathcal{C}_a(u, v)$ at $v = 0$ are $b_a(u)$ and $M_a(u)$. Hence the coefficient of v^2 in $\mathcal{C}_a(u, v)$ is $\mathcal{C}_a(u, 0)(b_a^2 + M_a)/2$. In the formal Gamma-moment factor, the coefficients of v^0, v^1, v^2 are respectively $\Gamma(1+u)$, $\Gamma''(1+u)/2$ and $\Gamma^{(4)}(1+u)/8$. Multiply these three finite coefficient expressions and use $\mathcal{C}_a(u, 0)\Gamma(1+u) = \mathcal{A}_a(u)$. This proves the identity without assuming analyticity in the full v variable. $\square$

In particular,

$$\begin{aligned} A_{(0,2)}(1) = \frac{\gamma_1^2 - \zeta''(2)}{2} + \frac{\gamma_1(\gamma^2 + \zeta(2))}{2} \\ + \frac{\gamma^4 + 6\gamma^2\zeta(2) + 3\zeta(2)^2 + 8\gamma\zeta(3) + 6\zeta(4)}{8}. \end{aligned} \quad (7.3)$$

Unlike the single-decoration constants at depth greater than one, this expression retains γ_1 . The comparison with (6.14) is useful: placing two logarithms on one index and placing them on two distinct indices are analytically different operations, even though their total logarithmic degree is the same.

The master formula gives every higher content by a finite recurrence. If $\alpha \neq 0$, $k = |\alpha|$, and $w = \sum_r r\alpha_r$, the coefficient of $\mathbf{u}^\alpha$ in $\log \mathcal{C}_a$ is

$$\lambda_\alpha = \begin{cases} \gamma_r(a), & \alpha = \mathbf{e}_r, \\ \frac{(-1)^{k-1+w}(k-1)!}{\prod_r \alpha_r!} \zeta^{(w)}(k, a), & k \geq 2. \end{cases} \quad (7.4)$$

For cutoff polynomials replace the singleton coefficient by $\gamma_r(a) + L^{r+1}/(r+1)$. If $F = \exp(\sum_{\alpha \neq 0} \lambda_\alpha \mathbf{u}^\alpha) = \sum_{\mathbf{m}} f_{\mathbf{m}} \mathbf{u}^{\mathbf{m}}$, then

$$|\mathbf{m}| f_{\mathbf{m}} = \sum_{0 < \alpha \leq \mathbf{m}} |\alpha| \lambda_\alpha f_{\mathbf{m}-\alpha}, \quad f_{\mathbf{0}} = 1. \quad (7.5)$$

This is obtained by applying $\sum_r u_r \partial_{u_r}$ to F . It supplies an exact finite algorithm, with no symbolic infinite sums and no integer-relation search.

8 Diagonal spectral finite parts are a different operation

For $\operatorname{Re} s > 1$ define

$$Z_d(s; a) = \sum_{n_1 > \dots > n_d \geq 0} \prod_{i=1}^d (n_i + a)^{-s}, \quad Z_0(s; a) = 1.$$

Newton's identity gives the coefficientwise meromorphic identity

$$\sum_{d \geq 0} Z_d(s; a) u^d = \exp \left(\sum_{k \geq 1} \frac{(-1)^{k-1}}{k} \zeta(ks, a) u^k \right). \quad (8.1)$$

For any fixed d , only $k \leq d$ is needed. Thus there is no need to define an infinite exponential on a punctured parameter domain in order to continue Z_d near $s = 1$.

Define the diagonal spectral constant

$$D_d(a) = \operatorname{CT}_{t=0} Z_d(1+t; a). \quad (8.2)$$

Here CT means the constant coefficient of a one-variable Laurent series. This is not, by definition, a cutoff or an Abel regularization.

Theorem 8.1 (Exact diagonal anomaly). *Let*

$$\begin{aligned} \mathcal{R}_a(u, t) = \exp \left\{ u \left(\zeta(1+t, a) - \frac{1}{t} \right) \right. \\ \left. + \sum_{k \geq 2} \frac{(-1)^{k-1}}{k} \zeta(k+kt, a) u^k \right\}. \end{aligned} \quad (8.3)$$

The expression is understood as a power series in u with coefficients holomorphic in t near zero. Then

$$D_d(a) = \sum_{j=0}^d \frac{1}{j!} [u^{d-j} t^j] \mathcal{R}_a(u, t). \quad (8.4)$$

The $j = 0$ term is $C_{(d)}(a)$. The other terms give a finite explicit correction involving Stieltjes constants only through index $d-1$ and spectral derivatives of Hurwitz zeta at positive integers.

Proof. Equation (8.1) gives $Z_d(1+t; a) = [u^d]e^{u/t}\mathcal{R}_a(u, t)$. Expanding $e^{u/t}$ and taking the constant Laurent coefficient gives (8.4). At $t = 0$, the regular factor is exactly $\mathcal{C}_a(u, 0, \dots)$, so the $j = 0$ coefficient is the cutoff constant. A Stieltjes coefficient of order j is accompanied by at least one u ; hence the largest possible index in the extraction is $d - 1$. Everything is a finite coefficient calculation once d is fixed. $\square$

For $g_r = \gamma_r(a)$ the first cases are

$$\begin{aligned} D_1(a) &= g_0, \\ D_2(a) &= \frac{g_0^2 - \zeta(2, a)}{2} - g_1, \end{aligned} \tag{8.5}$$

$$D_3(a) = \frac{g_0^3 - 3g_0\zeta(2, a) + 2\zeta(3, a)}{6} - g_0g_1 + \frac{g_2}{4} - \zeta'(2, a). \tag{8.6}$$

For example, $Z_2(s; a) = (\zeta(s, a)^2 - \zeta(2s, a))/2$. The product of the pole $1/t$ with the linear coefficient $-g_1t$ contributes $-g_1$ to its constant term. This contribution is absent in a cutoff constant extracted before collapsing the spectral variables.

At $a = 1$, the three unweighted schemes already differ in depth two:

scheme and normalization	depth-two constant
sharp cutoff, polynomial in $\log N$	$(\gamma^2 - \zeta(2))/2$
Abel endpoint, polynomial in $-\log(1-z)$	0
diagonal spectral Laurent constant	$(\gamma^2 - \zeta(2))/2 - \gamma_1$

(8.7)

These are different well-defined operations. In particular, the diagonal constant must not be called the standard stuffle-regularized value without an explicit change-of-regularization formula.

8.1 Independent variables and general rays

For absolutely convergent orders define the symmetrized shifted multiple zeta function

$$\mathcal{Z}(\mathbf{s}; a) = \sum_{\sigma \in S_d} \sum_{n_1 > \dots > n_d \geq 0} \prod_{i=1}^d (n_i + a)^{-s_{\sigma(i)}}.$$

The partition identity (3.10) gives its meromorphic continuation near $\mathbf{s} = \mathbf{1}$:

$$\mathcal{Z}(\mathbf{s}; a) = \sum_{\pi \in \Pi_d} \mu(\pi) \prod_{B \in \pi} \zeta\left(\sum_{i \in B} s_i, a\right). \tag{8.8}$$

Only singleton blocks have a pole near this point. If the variables $t_i = s_i - 1$ remain independent, the multivariate constant coefficient of $\prod_i (-\partial_{s_i})^{r_i} \mathcal{Z}$ equals the cutoff constant of the labelled symmetric harmonic sum. This follows block by block: a singleton has constant $\gamma_{r_i}(a)$, while a nonsingleton block has constant $(-1)^{\sum_{i \in B} r_i} \zeta(\sum_{i \in B} r_i)(|B|, a)$. Since the blocks have disjoint variables, there is no hidden cross-block Laurent contribution. Divide by $\prod_j m_j!$ to recover content normalization.

The situation changes upon restricting to a ray $t_i = c_i t$, where $c_i \neq 0$. This restriction is made *after* the independent spectral derivatives. For a singleton with decoration r_i ,

$$(-1)^{r_i} \zeta^{(r_i)}(1 + c_i t, a) = \frac{r_i!}{c_i^{r_i+1} t^{r_i+1}} + \sum_{m \geq 0} \frac{(-1)^m \gamma_{r_i+m}(a) c_i^m}{m!} t^m. \tag{8.9}$$

For a block B of size at least two, put $R_B = \sum_{i \in B} r_i$ and $c_B = \sum_{i \in B} c_i$.

Proposition 8.2 (Finite formula on an arbitrary nonsingular ray). *For a partition π , let $S(\pi)$ be its singleton indices and put $q_\pi = \sum_{i \in S(\pi)} (r_i + 1)$. Define*

$$T_i(t) = \frac{r_i!}{c_i^{r_i+1}} + t^{r_i+1} \sum_{m \geq 0} \frac{(-1)^m \gamma_{r_i+m}(a) c_i^m}{m!} t^m.$$

The ray constant of the labelled differentiated symmetric sum is

$$\sum_{\pi \in \Pi_d} \mu(\pi) [t^{q_\pi}] \left\{ \prod_{i \in S(\pi)} T_i(t) \prod_{\substack{B \in \pi \\ |B| \geq 2}} (-1)^{R_B} \zeta^{(R_B)}(|B| + c_B t, a) \right\}. \quad (8.10)$$

Only Taylor coefficients up to q_π are required in each summand.

Proof. In each term of (8.8), differentiation groups all spectral derivatives belonging to a block into the derivative of that block's single Hurwitz factor. Multiply each singleton expansion (8.9) by t^{r_i+1} . The whole partition contribution has then been multiplied by t^{q_π} . Its original Laurent constant is exactly the coefficient displayed in (8.10). $\square$

For example, the unweighted symmetric double has ray constant

$$g_0^2 - \zeta(2, a) - \gamma_1(a) \left(\frac{c_1}{c_2} + \frac{c_2}{c_1} \right). \quad (8.11)$$

Half of this at $c_1 = c_2 = 1$ is (8.5). Thus finite-part extraction and restriction to a spectral diagonal do not commute. This elementary counterexample is useful when moving an identity between a multivariate Hurwitz-jet calculation and a one-variable spectral specialization.

9 Exact antiderivatives and beta-weighted moment identities

This section gives convergent identities, with no endpoint finite-part assignment. Two different integration variables are useful. Integration against dz/z raises the outer polylogarithmic index. Ordinary integration against dz , or beta weights in $1-z$, has a different and particularly simple all-depth structure.

9.1 A kernel representation in the endpoint logarithm

Put $L = -\log(1-z)$. Expanding (5.5) in u and using $v = -\log(1-t)$ gives, for every $d \geq 1$,

$$E_{(d)}(z; a) = \frac{1}{(d-1)!} \int_0^L (L-v)^{d-1} (1-e^{-v})^{a-1} dv. \quad (9.1)$$

This identity also has a direct proof: the coefficient of u^d in $u(1-z)^{-u} B_z(a, u)$ is

$$\frac{1}{(d-1)!} \int_0^z \frac{t^{a-1}}{1-t} \left(\log \frac{1-t}{1-z} \right)^{d-1} dt,$$

and the stated change of variables yields (9.1). Every integral converges, since the kernel behaves as v^{a-1} near zero.

Consequently $E_{(d)}(1-e^{-L}; a)$ is the d -fold zero-based ordinary antiderivative of $(1-e^{-L})^{a-1}$ in the L variable. In particular, $\partial_L E_{(d)} = E_{(d-1)}$ for $d \geq 2$, while $\partial_L E_{(1)} = (1-e^{-L})^{a-1}$. This is an exact full-function statement, not merely an endpoint asymptotic.

Theorem 9.1 (Beta-weighted all-depth moments). *For $a, b > 0$ and integers $d \geq 1$,*

$$\boxed{\int_0^1 (1-z)^{b-1} E_{(d)}(z; a) dz = \frac{B(a, b)}{b^d}.} \quad (9.2)$$

For any integer $k \geq 0$, one may differentiate k times in b :

$$\int_0^1 (1-z)^{b-1} \log^k(1-z) E_{(d)}(z; a) dz = \frac{B(a, b)}{b^d} Y_k(\Delta_1, \dots, \Delta_k), \quad (9.3)$$

where $Y_0 = 1$ and

$$\Delta_j = \psi^{(j-1)}(b) - \psi^{(j-1)}(a+b) + \frac{(-1)^j d(j-1)!}{b^j}, \quad j \geq 1. \quad (9.4)$$

Proof. With $L = -\log(1-z)$ the left side of (9.2) is $\int_0^\infty e^{-bL} E_{(d)}(1-e^{-L}; a) dL$. The integrand in (9.1) is nonnegative, so Tonelli's theorem allows the integration order to be changed. The inner Gamma integral is

$$\frac{1}{(d-1)!} \int_v^\infty e^{-bL} (L-v)^{d-1} dL = \frac{e^{-bv}}{b^d}.$$

The remaining integral, after $t = e^{-v}$, is $\int_0^1 t^{b-1} (1-t)^{a-1} dt = B(a, b)$. This proves the moment formula. The integral and every fixed b derivative converge uniformly for b in a compact positive interval: the added factor is a power of L , while the Laplace damping remains exponential. Hence differentiation under the integral is valid. The logarithm of the right-hand side has j th derivative Δ_j , so the complete Bell-polynomial derivative rule gives (9.3). $\square$

Several particularly transparent identities follow:

$$\int_0^1 E_{(d)}(z; a) dz = \frac{1}{a}, \quad d \geq 1, \quad (9.5)$$

$$\frac{1}{(q-1)!} \int_0^1 (1-z)^{q-1} E_{(d)}(z; a) dz = \frac{1}{(a)_{q,q^d}}, \quad q \in \mathbb{N}, \quad (9.6)$$

$$\int_0^1 z^{h-1} E_{(d)}(z; a) dz = \sum_{j=0}^{h-1} (-1)^j \binom{h-1}{j} \frac{B(a, j+1)}{(j+1)^d}, \quad h \in \mathbb{N}. \quad (9.7)$$

The last identity follows by expanding $z^{h-1} = (1 - (1-z))^{h-1}$. It is a finite exact evaluation, not an asymptotic expansion. For example,

$$\int_0^1 z E_{(d)}(z; a) dz = \frac{1}{a} - \frac{1}{a(a+1)2^d}.$$

A reflection specialization is especially simple. For $0 < a < 1$,

$$\int_0^1 (1-z)^{-a} E_{(d)}(z; a) dz = \frac{\pi}{\sin(\pi a)(1-a)^d}. \quad (9.8)$$

For example, put $\ell_2 = \log 2$. At the half lattice,

$$\begin{aligned} \int_0^1 \frac{E_{(d)}(z; 1/2)}{\sqrt{1-z}} dz &= 2^d \pi, \\ \int_0^1 \frac{\log(1-z) E_{(d)}(z; 1/2)}{\sqrt{1-z}} dz &= -2^{d+1} \pi (d + \ell_2), \\ \int_0^1 \frac{\log^2(1-z) E_{(d)}(z; 1/2)}{\sqrt{1-z}} dz &= 2^d \pi \left(4(d + \ell_2)^2 + 4d + \frac{\pi^2}{3} \right). \end{aligned} \quad (9.9)$$

These follow from $B(1/2, 1/2) = \pi$, $\psi(1/2) - \psi(1) = -2\log 2$, and $\psi'(1/2) - \psi'(1) = \pi^2/3$. The family is uniform in the depth; the logarithms here decorate the integration weight, and should not be confused with spectral logarithms at the lattice indices.

For positive integers $a = M$, $b = N$, formula (9.4) becomes

$$\Delta_j = (-1)^j(j-1)! \left(H_M^{(j)}(N) + \frac{d}{N^j} \right). \quad (9.10)$$

Thus every logarithmic moment in (9.3) at such parameters is an explicit rational expression in finite generalized harmonic numbers, with a rational beta prefactor. The simple moment laws are special-function consequences of the kernel representation; no priority is claimed for their unshifted log-power specializations.

Corollary 9.2 (Gamma-weighted generalized harmonic sums). *For $a, b > 0$ and $d \geq 1$,*

$$\sum_{n \geq 0} e_{(d-1),n}(a) \frac{\Gamma(n+a)}{\Gamma(n+a+b+1)} = \frac{\Gamma(a)}{b^d \Gamma(a+b)}. \quad (9.11)$$

In particular,

$$\begin{aligned} \sum_{n \geq 0} \frac{\Gamma(n+a) H_n^{(1)}(a)}{\Gamma(n+a+b+1)} &= \frac{\Gamma(a)}{b^2 \Gamma(a+b)}, \\ \sum_{n \geq 0} \frac{\Gamma(n+a) ((H_n^{(1)}(a))^2 - H_n^{(2)}(a))}{\Gamma(n+a+b+1)} &= \frac{2\Gamma(a)}{b^3 \Gamma(a+b)}. \end{aligned} \quad (9.12)$$

Proof. Condition on the largest index in the beta moment. Tonelli's theorem gives

$$\int_0^1 (1-z)^{b-1} E_{(d)}(z; a) dz = \sum_{n \geq 0} \frac{e_{(d-1),n}(a)}{n+a} B(n+a+1, b).$$

Write each beta factor as a Gamma quotient, use Theorem 9.1, and cancel $\Gamma(b)$. The two displayed examples use $e_{(1),n} = H_n^{(1)}$ and $2e_{(2),n} = (H_n^{(1)})^2 - H_n^{(2)}$. $\square$

For an explicit half-lattice specialization, let $\text{Cat}_n = \binom{2n}{n}/(n+1)$ be the Catalan numbers. Since

$$\frac{\Gamma(n+1/2)}{\Gamma(n+2)} = \sqrt{\pi} \frac{\text{Cat}_n}{4^n}, \quad H_n^{(j)}(1/2) = 2^j H_{2n}^{(j)} - H_n^{(j)},$$

we obtain the convergent harmonic identities

$$\sum_{n \geq 0} \frac{\text{Cat}_n}{4^n} (2H_{2n} - H_n) = 4, \quad (9.13)$$

$$\sum_{n \geq 0} \frac{\text{Cat}_n}{4^n} ((2H_{2n} - H_n)^2 - 4H_{2n}^{(2)} + H_n^{(2)}) = 16. \quad (9.14)$$

More generally, the corresponding sum of $e_{(d-1),n}(1/2)$ is 2^d . These identities are beta/Gauss consequences of the unweighted generator, included to make the finite generalized-harmonic content explicit; no claim that the individual low-depth Catalan specializations are previously unknown is intended.

9.2 Ordinary zero-based primitives and integer-shift elementary forms

Define the q -fold ordinary primitive by

$$I_{q,d}(z; a) = \frac{1}{(q-1)!} \int_0^z (z-t)^{q-1} E_{(d)}(t; a) dt, \quad q \geq 1.$$

Termwise integration of (5.4) gives

$$\sum_{d \geq 1} I_{q,d}(z; a) u^d = \frac{u \Gamma(a) z^{a+q}}{\Gamma(a+q+1)} {}_2F_1(1, a+u; a+q+1; z). \quad (9.15)$$

At $z = 1$, Gauss summation reduces the right-hand side to $u/((a)_q(q-u))$, agreeing with (9.6) coefficient by coefficient. Either the beta-moment proof or the convergent Gauss formula justifies this endpoint evaluation for u near zero.

At a positive integer shift the kernel in (9.1) is a finite exponential polynomial. Therefore the full nested harmonic function is elementary:

$$E_{(d)}(z; M) = \frac{L^d}{d!} + \sum_{j=1}^{M-1} \frac{(-1)^j \binom{M-1}{j}}{(-j)^d} \left((1-z)^j - \sum_{k=0}^{d-1} \frac{(-jL)^k}{k!} \right), \quad L = -\log(1-z). \quad (9.16)$$

To prove this, expand $(1 - e^{-v})^{M-1} = \sum_{j=0}^{M-1} (-1)^j \binom{M-1}{j} e^{-jv}$ in (9.1) and integrate each exponential d times with zero initial conditions. The $j = 0$ term gives $L^d/d!$; the Taylor subtraction in the other terms enforces those initial conditions exactly.

For $M = 2$ the first two cases are

$$E_{(1)}(z; 2) = L - z, \quad E_{(2)}(z; 2) = \frac{L^2}{2} - L + z.$$

Dropping the exponentially small $(1-z)^j$ terms in (9.16) gives precisely the endpoint polynomials from (5.9). Equating their constant coefficients also recovers the finite symmetric-function identity

$$h_d(1, 1/2, \dots, 1/(M-1)) = \sum_{j=1}^{M-1} \frac{(-1)^{j-1} \binom{M-1}{j}}{j^d}, \quad M \geq 2,$$

where h_d is the complete homogeneous symmetric function. This finite identity is another classical specialization, not an independence statement.

9.3 Logarithmic primitives raise the outer polylogarithmic index

Let $D_z = z \partial_z$. For a nonempty content and $p \geq 1$, put

$$J_{p,\mathbf{m}}(z; a) = \frac{1}{(p-1)!} \int_0^z \left(\log \frac{z}{t} \right)^{p-1} E_{\mathbf{m}}(t; a) \frac{dt}{t}. \quad (9.17)$$

Then $D_z J_{p,\mathbf{m}} = J_{p-1,\mathbf{m}}$, where $J_{0,\mathbf{m}} = E_{\mathbf{m}}$, and all these primitives vanish at $z = 0$. Termwise integration gives

$$J_{p,\mathbf{m}}(z; a) = \sum_{n_1 > \dots > n_d \geq 0} \frac{z^{n_1+a}}{(n_1+a)^p} \sum_{\text{cont}(\mathbf{r})=\mathbf{m}} \prod_i f_{r_i}(n_i + a). \quad (9.18)$$

For $z = 1$ this is absolutely convergent. To see this, sum the absolute inner factors first. If the outer decoration is r , the inner coefficient is $O((1 + \log n)^{D-r-1})$ by elementary bounds on

harmonic sums. The outer factor is therefore $O((1 + \log n)^{D-1}/n^{p+1})$, a summable sequence. This also justifies the endpoint integral in (9.17).

For unweighted content (d) , the full generating function is

$$\sum_{d \geq 1} u^d J_{p,(d)}(z; a) = \frac{uz^a}{a^{p+1} p+2} F_{p+1} \left(\underbrace{1, a+u, \overbrace{a, \dots, a}^p}_{\underbrace{a+1, \dots, a+1}_{p+1}}; z \right). \quad (9.19)$$

Indeed, condition on the largest lattice point as in Proposition 5.2, and multiply its coefficient by $(n+a)^{-p}$. The ratio $(a)_n/(a+1)_n = a/(a+n)$ supplies each extra denominator. At $z=1$ the hypergeometric excess is $p-u$, so the expression is convergent for $\operatorname{Re} u < p$ in its usual parameter domain; a neighborhood of zero is enough for every coefficient identity used here.

9.4 All positive integer shifts: a finite Gamma deletion theorem

For $M, p, d \in \mathbb{N}$, abbreviate

$$J_{p,d}(M) = J_{p,(d)}(1; M) = \sum_{n_1 > \dots > n_d \geq M} \frac{1}{n_1^{p+1} n_2 \dots n_d}, \quad P_{M-1}(u) = \prod_{j=1}^{M-1} \left(1 + \frac{u}{j} \right).$$

Theorem 9.3 (Integer-shift convergent antiderivative hierarchy). *As an identity of analytic germs at $(u, v) = (0, 0)$,*

$$\boxed{\sum_{p,d \geq 1} J_{p,d}(M) v^p u^d = \frac{1}{P_{M-1}(u)} \left(1 - \frac{\Gamma(1-u)\Gamma(1-v)}{\Gamma(1-u-v)} - \sum_{n=1}^{M-1} \frac{(u)_n}{n!} \frac{v}{n-v} \right)}. \quad (9.20)$$

Every coefficient belongs to the $\mathbb{Q}$ -algebra generated by ordinary zeta values. In particular,

$$\sum_{d \geq 1} J_{1,d}(M) u^d = \frac{-\psi(1-u) - \gamma - \sum_{n=1}^{M-1} (u)_n / (n!n)}{P_{M-1}(u)}. \quad (9.21)$$

Proof. The generating coefficient conditioned on the largest integer $n \geq M$ is

$$\sum_{d \geq 1} \frac{u^d}{n^{p+1}} e_{d-1}(1/M, \dots, 1/(n-1)) = \frac{1}{P_{M-1}(u)} \frac{(u)_n}{n! n^p}.$$

Thus summing p gives the absolutely convergent series, locally near zero,

$$\frac{1}{P_{M-1}(u)} \sum_{n \geq M} \frac{(u)_n}{n!} \frac{v}{n-v}. \quad (9.22)$$

For example $|u|, |v| < 1/4$ suffices: the summands decay as $O(n^{-2+\operatorname{Re} u})$ uniformly on a smaller compact set.

For $M=1$ use the binomial identity $\sum_{n \geq 1} (u)_n t^n / n! = (1-t)^{-u} - 1$. Initially take $\operatorname{Re} v < 0$, $\operatorname{Re} u < 1$. Integration gives

$$\begin{aligned} \sum_{n \geq 1} \frac{(u)_n}{n!} \frac{v}{n-v} &= v \int_0^1 t^{-v-1} ((1-t)^{-u} - 1) dt \\ &= vB(-v, 1-u) + 1 \\ &= 1 - \frac{\Gamma(1-u)\Gamma(1-v)}{\Gamma(1-u-v)}. \end{aligned}$$

Both sides continue holomorphically near $(0, 0)$, so this proves the classical height-one Gamma identity there. Remove the finitely many terms $n < M$ in (9.22) to obtain (9.20). The Gamma logarithm (4.5) shows that its coefficients involve only ordinary zeta values and rational numbers; the finite sum and P_{M-1}^{-1} add only rational coefficients. Finally extract the coefficient of v to obtain (9.21). $\square$

For $M = 1$, equation (9.21) is the classical height-one identity $J_{1,d}(1) = \zeta(d+1)$. At $M = 2$ it gives

$$\begin{aligned} J_{1,1}(2) &= \zeta(2) - 1, \\ J_{1,2}(2) &= \zeta(3) - \zeta(2) + 1, \\ J_{1,3}(2) &= \zeta(4) - \zeta(3) + \zeta(2) - 1. \end{aligned} \tag{9.23}$$

The theorem covers all p and d , not only this first-primitive line. It does not assert that the general rational-shift hypergeometric expression (9.19), or its logarithmically decorated analogues, reduces to the same single-zeta algebra.

10 Rational shifts and cyclotomic transport

Let $a = h/q$ with $1 \leq h \leq q$, and write $\omega = e^{2\pi i/q}$ and $w = z^{1/q}$ for $0 < z < 1$. Define the common-shift ordered polylogarithmic sum

$$\mathcal{F}_{\mathbf{s}}(z; a) = \sum_{n_1 > \dots > n_d \geq 0} z^{n_1+a} \prod_i (n_i + a)^{-s_i}.$$

Proposition 10.1 (Root-of-unity filter with its normalization). *For $|z| < 1$ on a fixed root branch, and in particular on the positive real interval,*

$$\mathcal{F}_{\mathbf{s}}(z; h/q) = q^{s_1+\dots+s_d-d} \sum_{j_1, \dots, j_d=0}^{q-1} \omega^{-h(j_1+\dots+j_d)} \text{Li}_{\mathbf{s}}(w\omega^{j_1}, \omega^{j_2}, \dots, \omega^{j_d}). \tag{10.1}$$

The identity may be spectrally differentiated to any fixed order. In doing so, the factor $q^{s_1+\dots+s_d-d}$ must also be differentiated.

Proof. Put $k_i = qn_i + h$. All k_i are positive, strictly ordered and congruent to h modulo q . The denominators contribute $q^{s_1+\dots+s_d}$, and the outer exponential is w^{k_1} . Insert the exact filter

$$\mathbf{1}_{k \equiv h \pmod{q}} = \frac{1}{q} \sum_{j=0}^{q-1} \omega^{j(k-h)}$$

for each index and interchange the finite sums. This gives (10.1). Exponential damping makes the underlying sums and all fixed spectral derivatives locally normally convergent. Differentiation of the prefactor is necessary because $\log(n+a) = \log k - \log q$. $\square$

At the unweighted orders $s_i = 1$, the prefactor is one (q^0). For the logarithmic primitive with $s_1 = p+1$ and the other orders one, it is q^p . The latter formula remains absolutely convergent at $z = 1$ for $p \geq 1$, even after fixed spectral differentiations. Thus the convergent antiderivative results provide exact finite combinations of cyclotomic multiple-polylogarithm values, without any finite-part ambiguity. The beta moments similarly translate to identities involving root-of-unity polylogarithms and the Gamma ratio $B(h/q, b)$.

10.1 The logarithmic scale changes under root extraction

A separate adjustment is needed at a divergent Abel endpoint. With $z = e^{-\epsilon}$ and $w = e^{-\epsilon/q}$, the logarithmic endpoint variables are

$$L = \log(1/\epsilon), \quad T = \log(q/\epsilon) = L + \log q.$$

After spectral differentiation, let the *complete filtered sum* on the right of (10.1) represent $E_{\mathbf{m}}(z; h/q)$. Its polynomial in T is therefore

$$Q_{\mathbf{m}, h/q}(T - \log q). \quad (10.2)$$

The constant in the T normalization is $Q_{\mathbf{m}, h/q}(-\log q)$, whereas the constant in the original L normalization is $Q_{\mathbf{m}, h/q}(0)$. Their exact difference is

$$Q_{\mathbf{m}, h/q}(-\log q) - Q_{\mathbf{m}, h/q}(0) = \sum_{j=1}^{D(\mathbf{m})} \frac{(-\log q)^j}{j!} Q_{\mathbf{m}, h/q}^{(j)}(0). \quad (10.3)$$

This statement concerns the full finite trace and does not require a claim about independent asymptotic expansions of its individual colored summands.

For the smallest example, $a = 1/2$ gives

$$E_{(1)}(z; 1/2) = \text{Li}_1(\sqrt{z}) - \text{Li}_1(-\sqrt{z}) = \log \frac{1 + \sqrt{z}}{1 - \sqrt{z}}.$$

Its constant relative to $-\log(1 - z)$ is $2 \log 2$. Its constant relative to $-\log(1 - \sqrt{z})$ is only $\log 2$. The difference is not an error in either polylogarithm identity; it is a change in the specified endpoint scale. A cyclotomic coefficient compiler must preserve this distinction.

11 Reproducible coefficient computations and validation

The package implements the finite recurrence (7.5), not an unbounded symbolic expansion. A requested content bounds every auxiliary variable separately. The cutoff polynomial is computed first; the finite Gamma differential operator then gives the Abel polynomial. The same constant is independently recomputed by convolution with (4.7).

The symbols in the implementation have the following meanings:

Symbol	Meaning
$\mathbf{g0}, \mathbf{g1}, \dots$	$\gamma_0(a), \gamma_1(a), \dots$
$\mathbf{z2_1}, \mathbf{z3_2}, \dots$	$\zeta'(2, a), \zeta''(3, a), \dots$
$\mathbf{G}$	Euler's constant γ
$\mathbf{z2}, \mathbf{z3}, \dots$	Ordinary zeta values
$\mathbf{L}$	The explicitly specified endpoint logarithm

Treating these atoms as indeterminates is a device for exact polynomial verification. It does not assume that the corresponding numbers are algebraically or rationally independent.

11.1 Finite exact arithmetic

The exact suite compares direct finite products on rational alphabets with power-sum recurrences; checks the labelled set-partition formula; compares the two implementations of

Gamma transfer; verifies the unweighted cancellation, single- and double-decoration generators, higher single decorations, and the Gamma-conjugated differential recurrence; compares Newton and partition formulas for diagonal constants; verifies half and integer shifts; checks finite deletion identities for primitives and integer-shift elementary endpoint polynomials; and checks root-of-unity filters by exact remainders modulo cyclotomic polynomials.

The executed exact suite passed 2,600 assertions. Their grouping is recorded in `results/exact_checks.json` and summarized in the accompanying README. These are finite regression tests of the formulas and implementation. They do not replace the analytic proofs of normal convergence, asymptotic transfer, or interchange of infinite sums and integrals. No proof assistant was run.

11.2 Independent numerical checks

The executed numerical suite passed 158 comparisons at 60 decimal digits. The largest observed absolute residual was approximately 7.83×10^{-55} ; full values and tolerances are retained in `results/numerical_checks.json`. It compares finite decorated harmonic products with Stieltjes differences and Hurwitz-zeta derivatives; direct geometric-convergent sums with the hypergeometric and incomplete-beta generators; numerical integrals with the convergent primitive and beta-moment formulas; numerical parameter derivatives of Stieltjes constants with (5.13); and Cauchy averages of Laurent functions with the diagonal constants. It also checks the integer-shift elementary functions against direct harmonic coefficients.

These are ordinary floating-point computations, not interval enclosures. Reported residuals are observed residuals, not rigorous error bounds. In particular, a very small residual would not prove an identity that lacked its written proof. The exact suite and the numerical suite deliberately use different formulations where practical.

11.3 Direct endpoint diagnostics

A separate script sums the decorated Abel series directly and subtracts the *entire* predicted polynomial. It uses $N = \lceil 35/\epsilon \rceil$ terms, with $\epsilon = 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}$, for two shifts and four contents. At the smallest ϵ each sum uses 3,500,000 terms. The following observed differences at $a = 1$ illustrate why naive endpoint extrapolation can be misleading:

ϵ	$E_{(1,1)} - Q_{(1,1),1}$	$E_{(1,2)} - Q_{(1,2),1}$
10^{-2}	$1.6344939 \cdot 10^{-1}$	1.7682030
10^{-3}	$3.0587602 \cdot 10^{-2}$	$7.0392944 \cdot 10^{-1}$
10^{-4}	$5.0120208 \cdot 10^{-3}$	$2.0368075 \cdot 10^{-1}$
10^{-5}	$7.4952623 \cdot 10^{-4}$	$4.7670923 \cdot 10^{-2}$

Here Q is evaluated at $L = \log(1/\epsilon)$. The displayed differences contain the analytic remainder, finite truncation, and floating-point roundoff; they are not certificates for any of those components separately. Even millions of terms do not by themselves reveal many digits of the constant in a higher logarithmic content. The coefficient formulas, rather than extrapolated decimal values, determine the exact answers.

12 Targeted corrections and integration guidance

12.1 A persistent independence wording issue

In the reviewed snapshot, the canonical integration chapter, at the paragraph labelled `integral:neg:psim2`, describes a dependent PSLQ input basket and then says: “Basis atoms must be $\mathbb{Q}$ -independent.” The numerical diagnosis is useful, but the unrestricted requirement is misleading. Proving the needed independence is often a difficult open problem, and it is neither an input requirement for PSLQ nor an acceptable unstated assumption.

A precise replacement is:

Remove known exact rational dependencies from the candidate list, or work with representatives modulo the explicitly proved relation system. A relation finder may otherwise return an already-known relation that does not involve the target. No unproved rational independence of the remaining constants is assumed, and failure to find a target relation proves neither independence nor nonreducibility.

This is a correction to persistent wording, not a claim to have newly discovered the general PSLQ caveat. The manuscript’s broader distinction between numerical evidence and proof is retained. A replacement paragraph is supplied as an editorial fragment, not as an automatically applied patch.

12.2 Regularization conventions that should be stated explicitly

For any new endpoint or spectral identity, specify the limiting parameter, the logarithmic variable, and the subtraction polynomial. The explicit counterexample (8.7) shows why the bare phrase “finite part” is insufficient when changing between cutoffs, Abel damping, and a spectral diagonal. The scale law (10.3) supplies a second necessary convention when transporting to roots of unity.

These are proposed convention improvements and safeguards for integrating this continuation. We do *not* assert that a particular unreviewed incoming manuscript committed either mistake. Similarly, Proposition 4.3 is a warning about this article’s own generating series, not an allegation that the repository already treated it as holomorphic.

12.3 Recommended placement and status

A suitable self-contained report location is

`Analysis/Polylogarithms/docs/reports/endpoint-regularization/.`

The harmonic-depth discussion can point to the decorated coefficient calculus, while the integration and differentiation chapters can point to the Gamma-transfer, Stieltjes-derivative and primitive results. The report can also be cross-linked to the existing alternating-harmonic continuation, but their normalizations and research questions should not be merged merely because both use elementary symmetric harmonic functions.

The delivered files preserve a standalone article, proof status, runnable code and recorded outputs. The canonical manuscript should be edited only after independent mathematical review and comparison with the unexamined incoming payloads. This article proves the families stated here. It does not prove the repository’s S_6 or S_8 candidates, settle period independence, or establish minimum depth for any individual numerical value.

13 Further research questions

13.1 Ordered rather than fully symmetrized logarithmic values

The first obstruction beyond our theorem is already visible in depth two: separate the endpoint constants with numerators $\log n$ and $\log m$ rather than their sum. The present results determine their symmetric combination but not their difference. Determine a functional or integral representation for that antisymmetric component, then identify an exact basis of special functions sufficient for its endpoint constant. This is a reduction question, not a conjecture of arithmetic independence.

13.2 Convergent spectral decorations of the primitive hierarchy

Theorem 9.3 settles the unweighted integer-shift family at all outer indices and depths. With a logarithmic insertion, the series remains absolutely convergent after one dz/z integration, but the finite Gamma quotient no longer immediately evaluates the whole hierarchy. Determine which symmetrized insertions arise from derivatives of a finite-parameter Gamma or hypergeometric identity. The first target is the one-logarithm primitive at arbitrary depth, with the outer and inner spectral derivatives kept separate until the final symmetrization.

13.3 Mixed shifts and rational lattice relations

The common lattice is essential to the partition reduction. If each labelled slot has its own shift a_i , collision blocks involve products of distinct linear denominators rather than a single Hurwitz-zeta derivative. Partial fractions should give a finite extension for distinct shifts, with confluent limits producing polygamma derivatives. Establish a uniform formula that remains regular as shifts collide. For rational common shifts, use (10.1) and (10.3) to build explicit cyclotomic endpoint relations at small conductors without silently changing the finite part.

13.4 Which combinations eliminate positive-index Stieltjes atoms?

Theorem 6.1 explains a complete cancellation for one logarithmic decoration at $a = 1$. Two distinct logarithmic insertions retain γ_1 in (7.3). For a fixed total logarithmic degree, classify rational combinations of content constants whose coefficient polynomials eliminate all positive-index Stieltjes symbols. This can first be solved in the explicitly stated formal atom ring. Any subsequent claim about the independence of numerical periods would require additional proof.

13.5 Ray dependence and compatible renormalizations

Formula (8.10) reduces direction dependence to finitely many coefficients at each depth. Classify rays, or finite linear combinations of ray constants, that agree with independent-variable cutoff extraction through a prescribed depth. Depth two gives the explicit ratio $c_1/c_2 + c_2/c_1$, so even the first compatibility equation is nontrivial. A useful target is a normalization satisfying both finite-lattice deletion and cyclotomic scale transport, with its comparison to Abel regularization proved rather than stipulated.

13.6 Analytic meaning of the formal Gamma-moment series

The zero-radius result for $\mathcal{K}$ rules out a naive analytic master germ, but one-sided integrals can converge for suitable negative quadratic logarithmic parameters. Study sectorial or

distributional realizations whose asymptotic coefficients are exactly (4.7). Any such realization should preserve the already proved fixed-content identities and must specify its contour and parameter domain. No summability assertion is made here.

13.7 Exact moment transforms and formal verification

The beta-moment law diagonalizes the unweighted depth hierarchy under a Laplace transform in $-\log(1-z)$. Investigate whether analogous transforms exist for one or more spectral logarithmic decorations and whether they close under a finite polygamma differential system. Independently, the finite coefficient recurrence, the set-partition identity and the Gamma-conjugation rule are suitable first targets for Lean formalization. The analytic layer would then require formal proofs of the compensated product and the polynomial Laplace–Stieltjes transfer. The numerical suites in this package are not substitutes for those formal developments.

Conclusion

A single normally convergent harmonic product, followed by a finite Gamma operator, determines every cutoff and Abel polynomial in the stated symmetrized logarithmic family. The formulas expose precisely when positive-index Stieltjes constants survive and when they cancel. Diagonal spectral restriction introduces additional terms, and root extraction changes the endpoint scale; both effects have finite exact correction laws. On the convergent side, beta-weighted moments and ordinary primitives admit all-depth evaluations, while logarithmic primitives at integer shifts reduce to a Gamma quotient with an explicit finite deletion. These are proved identities with stated domains, not numerical reductions inferred from a small residual.

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- [7] ProveIt repository, *Canonical polylogarithms manuscript*, with editorial records in Analysis/Polylogarithms/docs/manuscript/, ProveIt repository. Reviewed snapshot: fc4d3bf80534ad7c901d3b8c9e71baf2df0064ed. <https://github.com/VladimirReshetnikov/ProveIt/tree/fc4d3bf80534ad7c901d3b8c9e71baf2df0064ed/Analysis/Polylogarithms/docs/manuscript>.
- [8] ProveIt repository, *Alternating harmonic polylogarithms*, report README and collection under Analysis/Polylogarithms/docs/reports/alternating-harmonic-polylogarithms/. README inspected at the same repository snapshot as [7]. <https://github.com/VladimirReshetnikov/ProveIt/tree/fc4d3bf80534ad7c901d3b8c9e71baf2df0064ed/Analysis/Polylogarithms/docs/reports/alternating-harmonic-polylogarithms>.
- [9] ProveIt repository, Incoming reports: intake and placement instructions, docs/incoming/README.md, same reviewed snapshot. <https://github.com/VladimirReshetnikov/ProveIt/blob/fc4d3bf80534ad7c901d3b8c9e71baf2df0064ed/docs/incoming/README.md>.